MATTER & METHOD

01

Building the same wall four ways. Feeling the difference.

CONCEPT

A tabletop installation. One wall section.
Four construction methods. One projector.

You physically build ONE method end-to-end, guided by projected light on the table. As each piece lands, data appears ON IT: CO■ cost, labor hours, supply chain origin.

The other three methods play as projected animations on adjacent zones — same table, no rebuilding, no visitor fatigue.

At the end: all four methods compared side by side. Same wall. Different worlds.

THE FOUR METHODS

MASONRY	3D PRINT	PREFAB	REUSED
Brick by brick	Layer by layer	Panel by panel	Salvaged brick
Traditional	Automated	Industrial	Circular

WHY REUSED BRICK?

Embodied carbon: near zero. Outperforms 3DP on every metric. Forces visitors to ask:
"If this is so much better, why don't we do it?"

The answer is policy, logistics, supply chains — not technology. That's the real conversation.

UX DECISION

Build ONE method. Animate three. Same comparison, no fatigue.

WHITE SPACE

No existing project combines interactive tabletop assembly + projection-mapped LCA data + multi-method comparison.

Refs: Gramazio Kohler Augmented Bricklaying (ETH) • Fologram Steampunk Pavilion (Tallinn 2019) • EPFL SXL reuse demonstrators • inFORM (MIT)

INTERACTION SYSTEM

Input → Processing → Output → Feedback loop

02

INPUT

Physical Placement

Overhead USB webcam reads ArUco fiducial markers on each piece.

No touch. No gesture. No wearable. The act of building IS the input.

Each marker encodes:
- Piece ID (which block)
- Position (x, y on table)
- Rotation (θ orientation)

PROCESSING

FSM + Validation

5-state finite state machine:
IDLE → READY → CHECKING
→ CONFIRMED / ERROR → COMPLETE

Validation per method:
- Masonry: ±5mm position
- 3DP: sequential layer order
- Prefab: ±10° rotation
- Reused: irregular — wider tol.

OUTPUT

Projection Feedback

GUIDE	Target outline for next piece
CONFIRM	Green pulse on valid placement
DATA	CO■ + hours + origin ON the piece
ERROR	Red outline + ghost correction
COMPARE	Final 4-method dashboard

FEEDBACK LOOP

Place piece → **Camera** detects → **FSM** validates → **Data projected** ON piece → **User reads** places next

← loop continues until all pieces placed — then method comparison triggers →

BONUS FEATURES

- Material-origin map ON each piece
- Carbon-budget clock ticking on table

• Material-origin map ON each piece • Carbon-budget clock ticking on table

- Failure states (misplace = animated collapse)
- LCA shown as ranges with sources
- [illegible]

SYSTEM & TIMELINE

03

Tech stack, key references, and term schedule

TECH STACK

Rhino / Grasshopper	Geometry + LCA data + assembly sequences
TouchDesigner	Runtime: camera, FSM, projection mapping
ArUco markers	Reliable tracking under projector light
OSC bridge	GH ↔ TD real-time communication
HD projector + webcam	Both overhead, closed feedback loop

WHY NOT PURE GRASSHOPPER?

GH is single-threaded. Anemone stalls canvas. Projector light contaminates webcam. TD handles CV + projection natively.

REFERENCES

Augmented Bricklaying AR-guided masonry for complex facades	Gramazio Kohler, ETH
Steampunk Pavilion Public assembly via holographic guidance	Fologram, Tallinn 2019
inFORM Canonical tangible data interaction	MIT Tangible Media 2013
EPFL SXL Reuse-driven structural pavilions	Corentin Fivet
Striatus Bridge 3D-printed concrete at structural scale	ZHA + BRG 2021
Material Cultures Bio-based construction comparisons	London 2021–

TERM TIMELINE